PRD — HARSHAL: Cinematic Personal Portfolio

Original Problem Statement

Build a premium, cinematic personal portfolio website for Harshal Vhatkar, conceived as a personal streaming-platform experience (not a Netflix clone, not a generic dev portfolio). Initial implementation covers the full entry experience only: (1) Hello handwriting animation, (2) HARSHAL identity reveal, (3) "Who's Watching?" profile selection (3 profiles), (4) shared featured hero with per-profile intro content. Later portfolio sections are explicitly out of scope for this phase.

User Personas

- **Visitor / curious friend** — explores the full "Meet Harshal" story.
- **Recruiter / hiring manager** — wants experience, education, skills, achievements fast.
- **Collaborator / client** — interested in product thinking, engineering and creative work.

Core Requirements (static)

- Palette: #090909 / #171719 / #B51F32 / #D33B4C / #E7E2DF / #A7A2A0; crimson used selectively as light, not surfaces.
- Typography: Dancing Script (hello only), Clash Display (display), DM Sans (body) — all self-hosted with licences kept in `public/licenses/`.
- Genuine handwriting reveal (SVG path stroke-draw), not typewriter/fade.
- Skip intro, once-per-session intro (user choice), browser back works, reduced-motion respected.
- Strict scope: no other sections, no fake stats, no stock portraits, no Netflix branding.

Architecture

- React (CRA + craco) SPA, react-router v7 with route-level AnimatePresence crossfades.
- Routes: `/` (hello → identity orchestrator), `/profiles`, `/profile/:profileId` (shared hero, config-driven).
- Content lives in `src/config/profiles.js`; design tokens in `src/index.css` + `tailwind.config.js`.
- Framer Motion for all sequencing; no new runtime dependencies added.
- Build-time asset pipeline: `scripts/generate_hello_paths.py` (fontTools + uharfbuzz) bakes real Dancing Script glyph contours (wght 500) into `src/data/helloPaths.js`.
- Backend untouched (FastAPI health only); no data layer needed this phase.

Implemented (2026-10-30)

- IntroAnimation: real stroke-drawn "Hello." (9 contours, staggered by stroke length, fill crossfade at ~2.9s, auto-advance).
- IdentityReveal: masked per-letter HARSHAL reveal → headline → roles → EXPLORE.
- EntryExperience: stage orchestrator, skip intro, sessionStorage once-per-session rule.

- ProfileSelection: 3 cards (monogram / typographic / abstract-node placeholders), hover glow + sibling recede, select focus → navigate.
- FeaturedHero: shared template, per-profile intro copy, masked line-by-line title, portrait stage with rim light + ghost monogram placeholder (swap via `portrait` field), pointer parallax, placeholder CTAs (toasts).
- Self-hosted fonts (6 woff2) + licences (OFL ×2, Fontshare FFL), "H." SVG favicon, film grain, ambient crimson glow.
- THE TOOLKIT projection section (2026-10-31): hand-drawn 35mm projector art (spinning reels + crank when on), SWITCH ON/OFF toggle, warm-white light cone clipped from the lens across a light-table contact sheet with drifting dust motes; five film cards show number/title/tagline until switched on; hover focuses the light (siblings dim); clicking expands a film full-width — the beam grows with the expanded film — revealing description, three capability columns and a BACK TO FULL VIEW control; programming-languages strip below; content stored in MongoDB (`toolkit_sections`, `toolkit_meta`) served by `GET /api/toolkit`, seeded via `backend/seed_toolkit.py`, with bundled fallback. Sections 1–5 content added; the focused per-film interior experiences are pending the user's next brief. Sizing pass (2026-10-31): camera enlarged/elongated, sheet and cards scaled up for full desktop, beam aperture widened plus a diffuse "screen wash" glow so the whole sheet reads illuminated (top of film 01 and bottom of film 04 included) while keeping realistic falloff.
- Expanded-film interior redesign (2026-10-31): fixed duplicate subtitle in expanded films (description paragraph only renders when a separate tagline exists); capabilities/methods are now a storyboard of numbered "Scene" mini film frames; technologies/tools run on a continuously scrolling 35mm film strip (perforations, frame separators, pauses on hover, animates only while expanded); applied work renders as a credits roll (film name + credit note, crimson slate borders). All five Revify screenshots received, converted to PNG and wired into the film-section gallery (sixth pending).
- CustomCursor (2026-10-30): crimson dot + trailing ring + lagging ambient glow follow the pointer; ring expands with a crimson tint over interactive elements, presses shrink it; native cursor hidden via `has-custom-cursor`; disabled for touch and reduced-motion users.
- OFF-SCRIPT post-credits section (2026-10-31): the personal-interests reel framed as post-credit scenes — eyebrow "POST-CREDITS SCENES // FOR THE ONES WHO STAYED", heading "OFF-SCRIPT." (crimson period) and a nudge subtitle acknowledging viewers who stayed through the credits. Four alternating editorial scene rows, each with its own hand-drawn SVG prop in the Toolkit projector art style that draws itself on scroll (pathLength) — clapperboard (Movies & Shows, "STILL WATCHING?" — snaps shut on hover), mic + keyboard with crimson sound waves (Music & Singing, "THE UNTRAINED VOICE", with a "Performance stills" strip showing graceful STILL PENDING placeholders for the user's future singing/keyboard photos at /offscript/stills/), editing timeline with crimson playhead (Video Editing, "CUT TO: EVERYTHING"), theatre masks with crimson smile/frown (Acting, "TWO YEARS ON STAGE"). Content data-driven via GET /api/offscript (`offscript_scenes`) seeded by `backend/seed_offscript.py`, with bundled fallback. Sits between The Filmography's "THE NEXT CHAPTER?" ending and the end-credits contact footer — the exact film-order experience. QA iteration_3.json: 100% — backend 6/6, alternating layouts, hover snap, pending stills, mobile stack, reduced-motion, zero regressions.

- Off-Script polish pass (2026-10-31, user feedback): slates compacted (visual column 0.8fr, prop art capped at 380px/300px mobile) so text reads dominant; props enriched with gradients, ground shadows, toolbar dots, rivets and grille/ring details; every prop gained quiet idle motion — clapper + keyboard bob, mic sound waves pulse with a crimson glow breathing behind the head, the timeline playhead scrubs across the cut while staggered audio EQ bars dance, masks drift on opposite phases — all CSS-driven, disabled under prefers-reduced-motion; slate gained viewfinder corner ticks that brighten crimson on hover; scene titles bumped to clamp(28px,2.6vw,40px). Verified via desktop + mobile screenshots (animations caught mid-motion, no overflow).

- YOUR NEXT WATCH 35mm reel rebuild (2026-11-01, user-approved plan): replaced the flat 4-face "cube" rotation with a continuous curved film band — each frame is windowed across 10 vertical facets placed on a true cylinder (radius derived from frame width so the band wraps gap-free; facet spans centred on frame angles), sprocket perforations and content flow continuously around the curve; reel physics on a rAF loop (drag tracks velocity → fling coasts with film-weight friction → eased settle with slight mechanical overshoot → clunk; arrows/dots/wheel/keyboard retarget the settle mid-motion; live counter + frame name during motion); per-facet shading overlay gives continuous depth-of-field (film falls into shadow away from the front); invisible per-frame probe panels feed the expansion FLIP origin; resize rebuilds geometry preserving rotation state. Synthesized sound design (new `ynwSound.js`, WebAudio, zero assets): reel whir with pitch/gain following spin speed, sprocket ticks on frame boundaries, clunk on settle, projection swell on frame expansion, collapse sweep on close, shutter click on lightbox zoom — unlocked on first gesture, SOUND ON/OFF chip in the section header (persisted). Header: crimson period after "YOUR NEXT WATCH." + margin-bottom 72px. QA iteration_4.json: 100% — geometry (40 facets/reel), arrows, fling momentum, click zones, expansion + gallery + lightbox, ESC, sound toggle persistence, resize 1920→390, mobile 390px, keyboard rotation (01→02→01), zero console errors, zero regressions.

- THE FILMOGRAPHY experience section (2026-10-31): cinematic editorial header ("THE FILMOGRAPHY." with crimson period) + three poster-style chapter cards — SPIT "The Foundation" (campus photo), SPJIMR "The Builder's Cut" (glowing-laptop photo), Baker Hughes "Behind the Scenes" (welder photo) — each with chapter label chip, org logo in a white chip, title/org/role/duration/description, "Explore Chapter" cue; hover = image zoom + restrained crimson tint + lift. Click FLIPs the card into a reusable expanded chapter detail (scrollable, backdrop dims+blurs the page, Back/ESC/backdrop close, scroll position restored) with meta grid, metric typography (SPJIMR 7+/20+/3/5+; Baker Hughes 180+/100+/×2/0) and numbered chapter highlights — facts sourced from the brief + the uploaded resume only. Final recruiter ending: dark band with "And now, the most Interesting part" / "THE NEXT CHAPTER?" / script-font "I'm looking right at it." / "Perhaps you're part of what comes next." + "LET'S MAKE SOMETHING HAPPEN" scrolling to the contact footer. Content data-driven via GET /api/filmography (`filmography_chapters`) seeded by `backend/seed_filmography.py`, with bundled fallback. End-credits contact footer at page end (#contact): H. wordmark, "A Mind Without a Genre", Email (mailto:harshal.v.official@gmail.com), LinkedIn, GitHub pills. QA iteration_2.json: 100% — 10/10 backend tests (backend/tests/test_filmography.py), desktop+mobile layouts, keyboard a11y, zero regressions.

- Toolkit cinematic projection pass (2026-11-01, user feedback): detailed machined camera rebuild in `CameraArt.js` (metal gradients, layered body panels, glass viewfinder, film thread with sprocket dashes, twin reels with crimson hubs, REC lamp, hood + hot lens glass); camera column lowered so

the lens sits at the contact sheet's vertical centre (padding-top 146px desktop / 106px ≤1024 / 0 ≤860) and the beam now fans as a full symmetric V; light re-architected into layers — a light-table pool renders UNDER the sheet content, only a whisper veil above it, and a travelling wedge (`.tk-beam--escape`, z-index -1 in the root context, clip-path measured from the lens, upper slope 0.62 / lower 1.05, warm gradient fading 0.09→0.03, blur 30px) escapes the section to slide BEHIND the neighbouring sections: faintly grazing Your Next Watch above and washing down into Filmography toward the Baker Hughes row while every card stays forward and readable (section roots .tk-section/.ynw-section/.fg-section/.os-section made transparent-floored, #root no longer paints a background — see fragility comment in toolkit.css). Wheel-rotation + lamp warm-up sounds via the shared WebAudio engine (toolkit-sound-toggle chip, cross-synced with YNW); sheet sized up again (max-width 1024, taller card faces, larger titles/language chips). QA iteration_5.json: 100% — power toggle, beam escape geometry (clip verified into both neighbours), 6 readable cards, camera not washed out, Filmography expand/collapse sound paths (back/ESC/backdrop), sound chip, YNW regression, zero console errors, no overflow 1440/390.

- Toolkit icons + languages film + resume (2026-10-31): tools film strip renders self-hosted brand glyphs (26 marks baked from Simple Icons path data in `toolIcons.js`, monochrome warm-white; Apify, Premiere Pro, Da Vinci Resolve, Capcut intentionally text-only); "Programming languages" moved into the empty 6th slot of the contact sheet as a static film card `tk-card--langs` (number 06, icon chips for C/C++/Java/JavaScript/Python, non-expandable, dims while a film expands — old strip below the sheet removed); hero VIEW RESUME now opens `ResumeChooser.js` — cinematic modal (film perfs, View Resume opens `/Harshal_Vhatkar_APM_Resume.pdf` in a new tab, Download PDF saves as `Harshal_Vhatkar_Resume.pdf`, closes via X/ESC/backdrop with scroll lock). Resume PDF served from `frontend/public/`. QA (testing agent, iteration_1.json): 100% pass — modal flows, strip icons 7/7 & 4/4, loops seamless, YNW gallery regression clean, cursor above modal, mobile 390px fits.

- YOUR NEXT WATCH film-roll section (2026-10-31): dual 3D cylinder reels (THE BUILDS / THE THINKING) ported from the user's prototype — drag, wheel, touch, arrows, dots, keyboard; shared-element "expansion portal" opening full case studies (Revify inline + EmFlow, Swiggy, Revify PRD, RAG Assistant/Evaluation modules reused verbatim); gallery lightbox; showcase tab/slider interactions added; deterministic click zones replace unreliable 3D hit-testing. Catalog (8 frames) stored in MongoDB `ynw_frames`, served by `GET /api/ynw/reels`, seeded via `backend/seed_ynw.py`, with a bundled fallback catalog. EXPLORE MY WORK now scrolls to the section. Revify gallery shows "SCREENSHOT PENDING" placeholders until the real screenshots land in `public/revify-project-screenshots/`.

- Polish pass (2026-10-31): custom cursor now renders above the YNW expansion portal and gallery lightbox (dot/ring z-index 1002/1001, hover state added to gallery cards); YOUR NEXT WATCH header left-aligned to THE TOOLKIT at every breakpoint (container widened to 1560px, mobile side paddings matched at 18px); toolkit tools film-strip now repeats its items so the translateX(-50%) loop stays seamless for short lists (Databases & Creative no longer run empty mid-scroll); 6th Revify screenshot ("Detailed Feature Analysis") wired into gallery slot 6 (`public/revify-project-screenshots/6.png`) with an updated Screen 06 caption. Verified: cursor visible over expanded frame + lightbox, header left edges identical (0px diff desktop & mobile), both strips loop seamlessly.

Backlog

- **P0**: Build "Meet Harshal" sections (story, projects, ideas) under the hero.
- **P1**: Upload the real singing-performance screenshots + keyboard photo into `frontend/public/offscript/stills/` (sing-1.png, keys-1.png) to replace the pending stills.
- **P1**: Recruiter extras beyond The Filmography (achievements/honours).
- **P2**: Real profile artwork to replace designed placeholders.

Known Limitations

- Handwriting animation draws each glyph's true contours (outline-draw that fills in) — the standard technique for real fonts; a single-line pen-stroke look would require a monoline script or hand-skeletonized paths.
- EXPLORE MY WORK scrolls to Your Next Watch; VIEW RESUME opens the chooser (view in new tab / download PDF). No further placeholder toasts remain.
- Intro is once per session by user choice; clear sessionStorage key `harshal:intro-seen` to replay.
- Tool icons are monochrome (palette-driven), not brand-colored, to keep the obsidian/crimson aesthetic.

Test Credentials

None — the experience has no auth or data layer yet.